

Math515 – Financial Mathematics II

Spring 2014

Homework 2

1. For a (standard) Brownian motion/Wiener Process W_t show that

$$E[W_t^p] = t^{p/2}(p-1)(p-3)\cdots 1$$

for even integers p .

2. Suppose that W_t is a Brownian motion. Define a stochastic process X_t by $X_t = tW_{1/t}$ for $t > 0$, and define $X_0 = 0$. Is X_t is also a Brownian motion?
3. Define a stochastic process X_t by $X_t = e^{-t}W_{e^{2t}}$ for $t \geq 0$. X_t is called an *Ornstein-Uhlenbeck* process. Find the covariance function of X_t .
4. The Gaussian process defined by

$$X_t = \frac{W_{t+h} - W_t}{h}, \quad t \geq 0,$$

with $h > 0$ being some fixed constant is called a “*colored noise*”. Show that

$$\mu_X(t) = 0 \quad \text{and} \quad c_X(s, t) = \frac{1}{h^2}[(s+h) - \min(s+h, t)], t \geq s \geq 0.$$

5. Estimate (albeit crudely) the stock drift and stock volatility of the price of silver (US\$/ounce) from the data given below:

Year	Closing Price
2000	\$4.595
2001	\$4.650
2002	\$4.790
2003	\$5.960
2004	\$6.845
2005	\$8.910

6. Modify the given MATLAB code of the Euler method to numerically solve and plot five different realization of the solutions of the following SDEs:

(a) $dX_t = \left[\frac{2X_t}{1+t} + (1+t)^2\right] dt + (1+t)^2 dW_t$ with $X_0 = 0$;

(b) $dX_t = dt + 2\sqrt{X_t}dW_t$ with $X_0 = 1$;

(c) $dX_t = -\frac{1}{2}e^{-2X_t}dt + e^{-X_t}dW_t$ with $X_0 = 1$;

(d) $dX_t = X_t^3 dt - X_t^2 dW_t$ with $X_0 = 0$;

The code "SDE_Euler.m" of the Euler method for solving SDEs:

```

%%*****
function SDE_Euler
%%Total simulation time [0,T]
T = 1;
%%Number of realizations/paths
npaths = 20;
%%Uniform time stepsize h=T/n
n = 1000;
t = linspace(0,T,n+1)';
h = diff(t(1:2));
%%Initialize X_0
x0 = 0;
x = x0*ones(n+1,npaths);
%%Solve by Euler method
for j = 1:n
    dw = randn(1,npaths)*sqrt(h);
    x(j+1,:) = x(j,:) + drift(t(j),x(j,:)).*h + volatility(t(j),x(j,:)).*dw;
end
%%Plot all realizations
plot(t,x)
xlabel('Time t');
ylabel('X_t');
title('Euler Method');
end

%%Define the drift term
function mu = drift(t,x)
mu = 1;
%mu = x;
%mu = 2*x/(1+t) + (1+t)^2;
end

%%Define the volatility term
function sigma = volatility(t,x)
sigma = 1.5;
%sigma = 1.5*x;
%sigma = 2*sqrt(x);
end
%%*****

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